

**Reproducing BioReason's LLM-Only Baselines
Locally:
Variant Pathogenicity Classification with Qwen3 on
Consumer Hardware**

Tasfia Ahammed Otry , 24301479, Section01, *CSE443*

Abstract

Interpreting genetic variants as either pathogenic (disease-causing) or benign (harmless) is an important challenge in genomic medicine. Modern sequencing methods can identify a very large number of genetic variants, making it difficult for clinical experts to manually classify all of them. This can delay the diagnosis of rare diseases and affect treatment decisions in precision oncology. This project reproduces the “LLM-only” baseline from the BioReason paper (Fallahpour et al., 2025) using smaller and more accessible Qwen3 language models. This study uses Qwen3-1.7B and Qwen3-4B locally through Ollama on a consumer GPU with 12GB of VRAM. An initial test was also performed using the larger Qwen3-8B model through cloud inference. Each model received a plain-text description of a genetic variant and was asked to reason step-by-step before predicting whether the variant was Pathogenic or Benign. The predictions were compared with ClinVar-based ground-truth labels from the VE-Non-SNV and VE-Coding benchmark datasets using accuracy, precision, recall, and F1-score. The results varied compared with the published baseline. On VE-Non-SNV, the smaller model achieved 58.42% accuracy compared with the paper’s 70.67%. However, on VE-Coding, the larger model achieved 77.50% accuracy, outperforming the paper’s 48.99%. The study also identified and fixed several practical problems, including failure to follow the prompt format, token-limit truncation, and prediction bias caused by class imbalance. Overall, this project provides a practical and transparent demonstration of how an LLM-based genetic variant classification system can be reproduced using more affordable hardware outside a research laboratory.

1 Introduction

Interpreting genetic variants is an important challenge in modern genomic medicine. A genetic variant is a change in DNA that may be either pathogenic, meaning it can cause disease, or benign, meaning it is harmless. A single human genome can contain around four to five million variants compared with the reference genome, but only a small number of these have been clearly classified by clinical experts, leaving many as variants of uncertain significance (VUS). Correctly classifying these variants matters because it affects the diagnosis of rare diseases, the selection of targeted cancer treatments, and decisions about screening for inherited diseases. As DNA sequencing becomes cheaper and more widely used, the number of variants requiring interpretation continues to grow, but there are not enough clinical experts to examine all of them manually — creating a need for automated methods that classify variants quickly, at scale, and ideally with explanations clinicians can check.

Several existing methods address this problem. The ACMG/AMP guidelines provide a structured, expert-driven framework for evaluating pathogenicity evidence but are not designed for automated large-scale processing [6], and much of the clinical evidence they rely on is drawn from ClinVar, a public archive of submitted variant interpretations [7]. Computational tools have automated parts of this process: CADD combines genomic and evolutionary features into a harmfulness score but does not explain its reasoning [5], while AlphaMissense uses deep learning over protein structure and evolutionary data to predict missense variant effects with strong performance, though it is limited to missense variants and likewise produces only a score [4]. DNA foundation models go further by learning patterns directly from sequence data — the Nucleotide Transformer produces representations useful for variant-effect prediction [2], and Evo 2, a 40-billion-parameter model trained on 9.3 trillion base pairs, extends this to genomes across life [3] — but both output embeddings or scores rather than human-readable explanations. Meanwhile, general-purpose LLMs such as the instruction-tuned Qwen family can perform step-by-step reasoning over text and run locally on consumer hardware [8], though it is unclear how much useful signal they can extract from a plain-text variant description without directly modeling DNA sequence. BioReason [1] attempts to bridge these strands by connecting a DNA foundation model with an LLM so the system can both predict and explain, reporting improved performance with interpretable reasoning — but reproducing its full architecture requires a specialized multimodal setup and over 24 GB of GPU memory, making it impractical for a typical student project or consumer hardware.

This leaves a gap between existing approaches: expert guidelines are reliable but not scalable [6], scoring tools are accurate but not interpretable [4, 5], DNA foundation models capture sequence patterns but not natural-language reasoning [2, 3], and BioReason offers both but at a computational cost few can reproduce [1]. This project therefore investigates a narrower question — how well a small, general-purpose LLM can classify genetic variants from a plain-text description alone, using only the "LLM-only" baseline configuration BioReason itself reports for Qwen3-1B and Qwen3-4B [1]. Rather than reproducing the full BioReason architecture, this work independently reproduces that LLM-only baseline on consumer-grade local hardware, and additionally surfaces practical issues — prompt-following failures, token-limit truncation, and class-imbalance-driven bias — that are not visible in the original paper's summary results.

1.1 Our Contribution

- Evaluated a larger cloud-hosted LLM (Qwen3-8B, accessed via a HuggingFace Inference Router endpoint on Google Colab) on a balanced 30-sample subset of the VE-Coding pathogenicity-classification split, establishing an initial baseline and validating the chain-of-thought prompting and answer-parsing approach before scaling to local, resource-constrained inference.
- Reproduced BioReason's published "LLM-only" baseline evaluation methodology end-to-end using local, open-weight Qwen3 models served through Ollama on a single consumer-grade 12GB-VRAM GPU, with no dependency on the original authors' or any cloud provider's compute.
- Benchmarked locally run Qwen3 models (1.7B and 4B parameter scales) against the paper's published Qwen3-1B/4B "LLM-only" rows on both the VE-Non-SNV and VE-Coding pathogenicity-classification splits.
- Built a reusable, parameterized inference-and-evaluation harness combining chain-of-thought prompting (reasoning trace plus fixed-format final answer), format-tolerant regex parsing of the model's verdict, and accuracy/precision/recall/F1 scoring against ClinVar-derived [7] ground-truth labels.
- Identified and corrected practical failure modes not visible in the original paper's summary table — including prompt-format non-compliance (the model copying an unfilled placeholder), local-inference truncation from token/context limits, and a measurable class-imbalance-driven prediction bias — relevant to anyone reproducing or deploying LLM-based variant classifiers outside a controlled research setting.

2. Material and Methods

Figure 1 shows the end-to-end pipeline used for every dataset in this study. A stratified sample is drawn from the HuggingFace test split, formatted into a fixed prompt template that asks the model to reason step-by-step and end with a single-line verdict, sent to a locally-hosted Qwen3 model through Ollama's OpenAI-compatible API, parsed with a regex-based extractor (with a fallback that scans the reasoning trace when the model fails to populate its final-answer field cleanly), and scored with scikit-learn against the dataset's ground-truth label. The same pipeline, with a cloud-hosted endpoint substituted for the local Ollama server, was used for the initial Qwen3-8B pilot run.

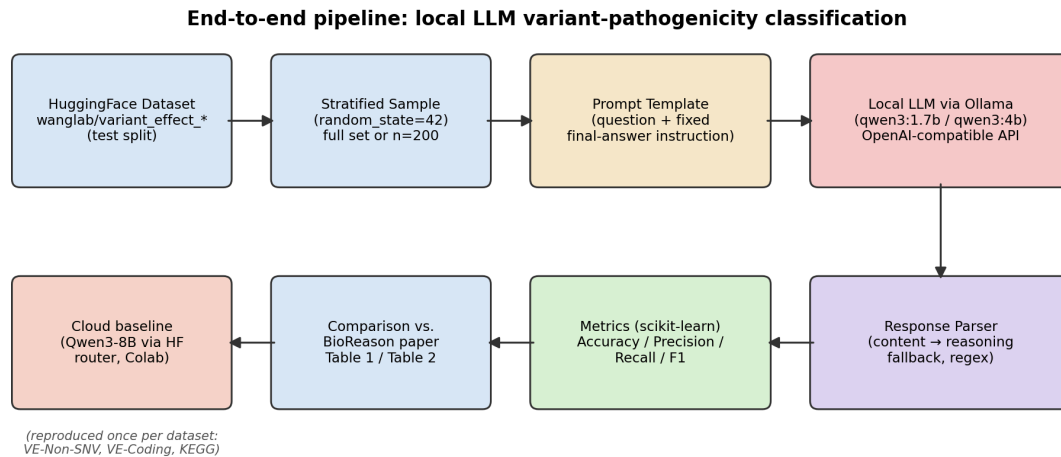


Figure 1. End-to-end pipeline for local LLM variant-pathogenicity classification.

2.1 Dataset Description

Two benchmark datasets from the BioReason paper were used for this project. They are part of the Variant Effect Prediction (VEP) evaluation and are available on Hugging Face under the wanglab organization. Both datasets are based on clinical variant information from ClinVar [7]. They contain a plain-text “question” field, which includes information such as the chromosome, position, and gene context, along with the correct pathogenic or benign label.

The VE-Non-SNV dataset already contains a “cleaned pathogenicity” column, which was used directly as the ground-truth label. In the VE-Coding dataset, the label was taken from the “answer” field. This field follows the format “Label; disease description”, such as “*Pathogenic; Neuronal ceroid lipofuscinosis 8*”. The label was obtained by splitting the field at the first semicolon.

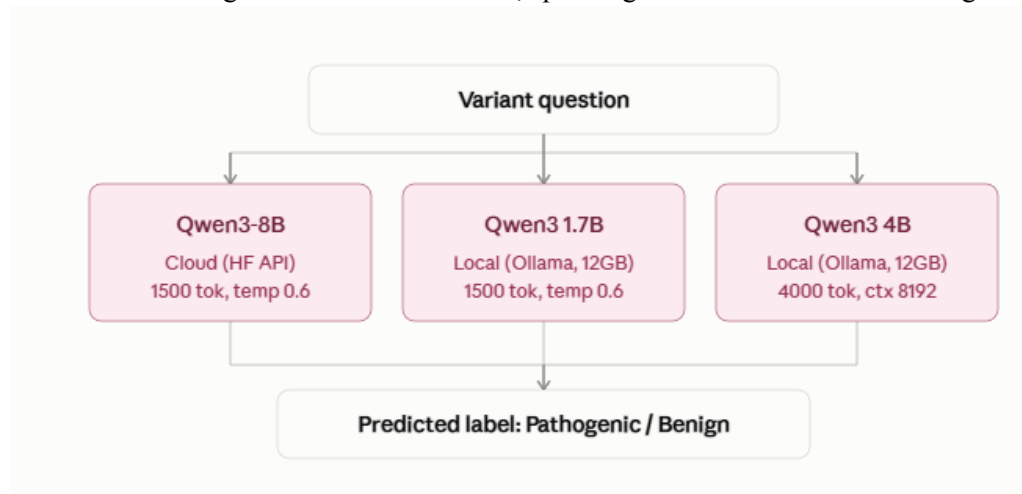
For this project, only the “question” field was given to the LLM as input. The DNA sequence and other reference columns were not used because they are required by the DNA-LLM fusion models, not by the LLM-only setup. This follows the same approach used for the LLM-only baseline in the BioReason paper [1].

The BioReason benchmark also includes a KEGG-based disease pathway reasoning dataset, which contains 37 classes and allows multiple labels. However, this dataset was not evaluated in the current phase of the project.

Dataset	HuggingFace ID	Test rows	Train rows	Classes	Test class balance
VE-Non-SNV	wanglab/variant_effect_non_snv	873	35,215	Pathogenic / Benign	523 / 350 (59.9% / 40.1%)
VE-Coding	wanglab/variant_effect_coding	1,233	48,850	Pathogenic / Benign	676 / 557 (54.8% / 45.2%)

2.2 Tools / Models / Algorithms Used

Three model configurations were evaluated, spanning both cloud and local hosting:



Model	Access	Hosting	max_tokens	num_ctx	temperature
Qwen3-8B	Cloud (HF Inference Router)	Google Colab	1500 (2500 on retry)	default	0.6
qwen3:1.7b	Local (Ollama)	12GB-VRAM consumer GPU	1500	default (~2048)	0.6
qwen3:4b	Local (Ollama)	12GB-VRAM consumer GPU	4000	8192 (explicit)	0.6

Three Qwen3 models were tested: Qwen3-8B through cloud inference and Qwen3-1.7B and Qwen3-4B locally using Ollama. All models received the variant “question” and were instructed to reason before giving the final answer as “Final Answer: [Pathogenic|Benign]” for automatic evaluation. The 4B model required a larger token and context limit because its longer reasoning was otherwise truncated. The models were selected to closely match the Qwen3-1B and Qwen3-4B LLM-only baselines reported in BioReason [1].

2.3 Performance Evaluation

This project uses zero-shot inference, meaning no model was trained or fine-tuned. The official test splits from each Hugging Face dataset, or a reproducible subset using random_state=42, were used

only for evaluation. Performance was measured using accuracy, precision, recall, and F1-score with scikit-learn.

Only Pathogenic and Benign predictions were considered valid. If the model produced an unclear or unrecognized answer (“Uncertain”), it was counted as incorrect rather than removed. Both macro-averaged and Pathogenic-class precision, recall, and F1-score were reported. The Pathogenic-class F1-score was also used because it was found to be closer to the results reported in the BioReason paper [1]. As no model training or fine-tuning was performed, no train/validation split was required; all data used was drawn from each dataset’s held-out test split for one-shot inference and scoring

3. Experimental Analysis

3.1 Cloud Pilot: Qwen3-8B on VE-Coding (n=30)

Before moving to local inference, a 30-row balanced sample (15 Pathogenic / 15 Benign) of VE-Coding was evaluated using Qwen3-8B via a cloud HuggingFace Inference Router endpoint, to validate the prompting and parsing approach. After excluding 3/30 (10.0%) responses that never resolved to a clean verdict even after a targeted retry, the remaining 27 rows scored as follows:

Class	Precision	Recall	F1	Support
Pathogenic	0.70	0.93	0.80	15
Benign	0.86	0.50	0.63	12
Overall accuracy	Na	Na	0.74	27

3.2 Local Reproduction: VE-Non-SNV and VE-Coding

Table 3 and Table 4 report accuracy, macro-averaged precision/recall/F1, and Pathogenic-class metrics for both local Qwen3 models on each dataset, alongside the paper’s published Qwen3-1B/4B “LLM-only” rows for direct comparison.

Table 3. VE-Non-SNV results

Model	n	Accuracy	Macro F1	Pathogenic F1	Uncertain rate
Paper — Qwen3-1B	873	70.67%	Na	76.21%	Na
Paper — Qwen3-4B	873	61.86%	Na	67.60%	Na
Ours — qwen3:1.7b	873	58.42%	45.98%	72.19%	1.1%
Ours — qwen3:4b	200	53.50%	53.36%	55.92%	0%*

Table 4. VE-Coding results

Model	n	Accuracy	Macro F1	Pathogenic F1	Uncertain rate
Paper — Qwen3-1B	1,233	46.55%	Na	34.82%	Na
Paper — Qwen3-4B	1,233	48.99%	Na	39.58%	Na

Model	n	Accuracy	Macro F1	Pathogenic F1	Uncertain rate
Ours — qwen3:1.7b	200	71.50%	74.28%	76.06%	13.5%
Ours — qwen3:4b	200	77.50%	77.48%	75.14%	0.5%

* All 53/200 (26.5%) initially-unresolved qwen3:4b/VE-Non-SNV responses were recovered by a reasoning-trace fallback parser that located the model's own already-stated verdict (see Section 3.3); 0% reflects the rate after this recovery step, applied consistently across all four local runs reported here.

Reproduced LLM-only baselines vs. BioReason paper (Table 2)

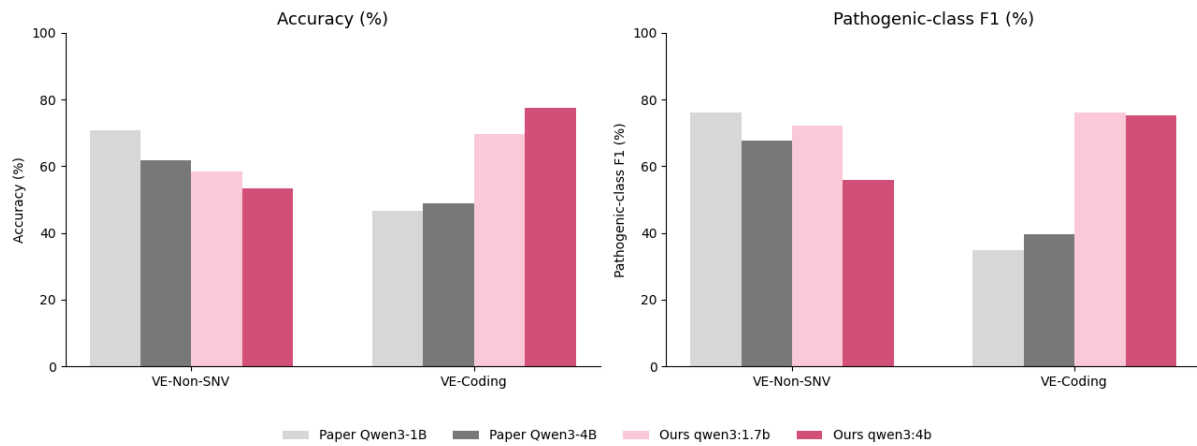


Figure 2. Reproduced accuracy and Pathogenic-class F1 vs. the BioReason paper's published Qwen3-1B/4B LLM-only rows, both datasets.

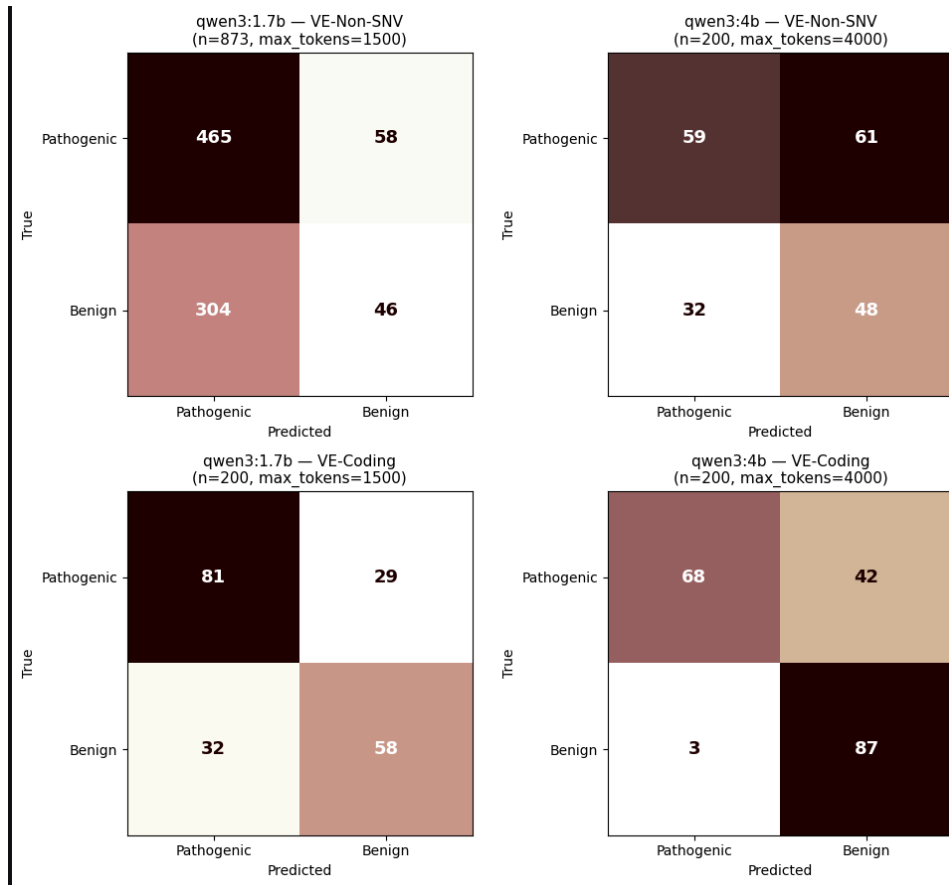


Figure 3. Confusion matrices for all four local runs (reconstructed from reported per-class precision/recall/support).

Two main patterns were observed. First, model size affected the two datasets differently. On VE-Non-SNV, the smaller qwen3:1.7b performed better than qwen3:4b (58.42% vs. 53.50% accuracy). This was also seen in the BioReason paper, where Qwen3-1B outperformed Qwen3-4B (70.67% vs. 61.86%) [1]. On VE-Coding, the larger model performed better in both cases: qwen3:4b achieved 77.50% compared with 69.50% for qwen3:1.7b, while the paper reported 48.99% for Qwen3-4B compared with 46.55% for Qwen3-1B [1].

Second, the models showed different prediction biases. On VE-Non-SNV, qwen3:1.7b predicted “Pathogenic” too often, with 89% recall for Pathogenic but only 13% for Benign. On VE-Coding, qwen3:4b was more conservative, with high Pathogenic precision (96%) but lower Pathogenic recall (62%). These results suggest that the prediction bias depends on the model, dataset, and prompt combination, rather than being a consistent behavior of the Qwen3 family.

3.3 Ablation Study: Token Budget and Context Window

An ablation was conducted to measure the effect of the generation and context settings on qwen3:4b. The initial configuration matched the settings used for qwen3:1.7b: max_tokens=1500 with Ollama’s default context window. On a fixed 200-row VE-Non-SNV sample, this configuration produced 194 unresolved responses (97%) because the model’s reasoning output was truncated before the required Final Answer line.

The configuration was then changed to max_tokens=4000 and num_ctx=8192, while keeping the model, prompt, dataset sample, and evaluation procedure unchanged. Under these settings, the same

200-row sample produced 0 unresolved responses, showing that the larger generation and context limits resolved the truncation problem. A preliminary 10-row test was used to verify the new configuration before running the full 200-row evaluation.

The change also increased inference cost. On the same 12GB-VRAM GPU, average inference time was approximately 4 seconds per row for qwen3:1.7b with a 1500-token limit and 32 seconds per row for qwen3:4b with a 4000-token limit and 8192-token context. Therefore, the 200-row sample was retained for qwen3:4b experiments because running the full test set would have substantially increased the computational time. This ablation demonstrates that generation and context limits can directly affect both output reliability and inference cost in local LLM evaluation.

3.4 Parsing Robustness

A second parsing issue was identified in the VE-Coding evaluation. In 12 of 200 qwen3:1.7b responses (6%), the model repeated the prompt placeholder “Final Answer: [Pathogenic|Benign]” instead of selecting a class. A simple regular expression could incorrectly interpret the first option, “Pathogenic,” as the prediction, creating a false increase in Pathogenic classifications.

To prevent this, the parser was modified to reject a candidate label when it was immediately followed by the “|” character. This prevented the placeholder from being treated as a valid prediction. Responses that still did not contain a clear Pathogenic or Benign decision were classified as “Uncertain” and scored as incorrect. A looser rule based on the last word of the reasoning was not used because manual inspection showed that these cases often contained genuine uncertainty rather than a clearly stated but misplaced prediction.

This approach prioritizes consistent and reproducible parsing over artificially increasing the model's score. The complete pipeline, including prompt templates, parsing logic, and evaluation code, is provided in the accompanying Jupyter notebooks for the VE-Non-SNV, VE-Coding, and Qwen3-8B cloud experiments.

<https://github.com/tassotry/variant-pathogenicity-llm-eval>

4. Conclusion

This project independently reproduced the “LLM-only” baseline from the BioReason paper using consumer-grade hardware. Qwen3-1.7B and Qwen3-4B were run locally through Ollama, with an initial Qwen3-8B cloud test used to validate the methodology. The results were mixed. The models performed worse than the paper's results on VE-Non-SNV, but performed much better on VE-Coding. The project also reproduced the paper's finding that the smaller model can perform better than the larger model on VE-Non-SNV [1].

In addition to the final performance scores, the project identified several practical issues that may not be visible in published results. These included token and context-limit truncation, incorrect copying of the answer placeholder, and prediction bias caused by class imbalance. Identifying and correcting these issues was important for obtaining reliable results when running LLMs on local hardware.

There are several limitations to consider. The local Qwen3 models were Ollama-quantized versions, so they may not be identical to the exact Qwen3-1B and Qwen3-4B checkpoints used in the BioReason paper. The original prompt used by the authors was also not available, meaning that the prompt used in this project was a reconstruction rather than an exact reproduction. In addition, the qwen3:4b experiments used a 200-row stratified sample, rather than the complete test sets of 873 rows for VE-Non-SNV and 1,233 rows for VE-Coding. This may introduce greater sampling variation. Finally, the KEGG disease-pathway reasoning dataset was not evaluated in this phase.

Future work will include evaluating the KEGG dataset using the same pipeline, running qwen3:4b on the full test sets if computational resources allow, and testing whether different token limits or more concise prompting can reduce reasoning-related truncation. Further research could build on BioReason's interpretable reasoning capabilities in three areas. First, a web-based variant interpretation system could allow researchers to upload variant data and view the model's reasoning in a structured dashboard. Second, BioReason could be evaluated on under-researched or region-specific diseases to investigate how well it performs on less-represented genetic conditions. Finally, newer systems such as BioReason-Pro could be explored for protein function and interaction prediction, extending the work beyond DNA variant classification.

References

- [1] Fallahpour, A., Magnuson, A., Gupta, P., et al. (2025). BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model. *arXiv preprint arXiv:2505.23579*.
- [2] Dalla-Torre, H., Gonzalez, L., Mendoza-Revilla, J., et al. (2025). Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22, 287–297.
- [3] Brixi, G., Durrant, M. G., Ku, J., et al. (2025). Genome modeling and design across all domains of life with Evo 2. *bioRxiv*.
- [4] Cheng, J., Novati, G., Pan, J., et al. (2023). Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, 381(6664), 1303–1310.
- [5] Rentzsch, P., Witten, D., Cooper, G. M., Shendure, J., & Kircher, M. (2019). CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Research*, 47(D1), D886–D894.
- [6] Richards, S., Aziz, N., Bale, S., et al. (2015). Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the ACMG and AMP. *Genetics in Medicine*, 17(5), 405–424.
- [7] Landrum, M. J., Lee, M. J., Benson, M., et al. (2016). ClinVar: public archive of interpretations of clinically relevant variants. *Nucleic Acids Research*, 44(D1), D862–D868.
- [8] Qwen Team. (2024). Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115*.
- [9] Wanglab. (2025). Variant effect datasets: variant_effect_non_snv and variant_effect_coding. HuggingFace Repository. <https://huggingface.co/wanglab>.